
DreamingGoose: Staged Distillation from Autoregressive Transformers to Bidirectional Recurrent Diffusion Language Models

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Abstract

Pretrained autoregressive Transformers embody enormous sunk compute. We ask whether that compute survives changing *both* the architecture (attention $\rightarrow$ recurrent state) and the generative objective (next-token $\rightarrow$ iterative denoising); prior conversion work crosses one axis at a time. We convert Qwen3 teachers at 1.7B and 8B into attention-free bidirectional gated-delta-rule diffusion students in three stages—attention-to-recurrence distillation, bidirectional expansion, and diffusion continual pretraining—so each capability can be traced to the stage that kept or lost it. Language modeling transfers. In-context retrieval does not: on a teacher-gated multi-query recall probe the converted students score 0.000 at both scales, and diffusion pretraining does not restore it. A table-to-query distance curriculum in the final stage repairs it, stochastically: under an open-loop schedule one seed in three retrieves. Making the schedule closed-loop—advancing retrieval distance only while a running accuracy estimate stays above threshold—lifts the same three seeds to 3/3 at 0.94–0.99, holds on real text (0.955–0.995 per cell), and carries unchanged to 8B (0.000 $\rightarrow$ 0.901 mean, generalizing to $2\times$ the trained load). The remaining 8B misses trace to a heavy-tailed bootstrap onset (6.5k, 11k, >16 k steps) that fixed step budgets truncate. One boundary survives every intervention: every solved model reads chance on held-out token bands, and a role-randomizing arm shows the gap is token *coverage*, not memorized bindings. We also complete a block-diffusion conversion of a 7B code model into a 3:1 recurrent–attention hybrid (85k steps, held-out NLL 2.819), with two negative training results.

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1. Introduction

Two conversion literatures exist and they do not intersect. One adapts pretrained autoregressive Transformers into *diffusion* language models while keeping the Transformer backbone (Gong et al., 2025; Ye et al., 2025). The other distills attention into *linear-time recurrence* while keeping next-token prediction (Goldstein et al., 2025). An attention-free diffusion LM built from a pretrained Transformer requires both moves, and to our knowledge no one has reported crossing both.

We do so in three stages, each with its own objective and its own set of frozen parameters (§3): distillation of the attention teacher into a causal recurrent student, expansion of that student into a bidirectional denoiser, and continual pretraining under a masked-diffusion objective. The staging is the method. When a capability fails to transfer, we can say at which stage it was lost.

The finding that organizes the paper is a dissociation. Converted students keep what the teacher *knows* and lose what the teacher can *do within a sequence*. On a multi-query associative recall probe (§3.4) where the attention teacher scores 0.335–0.350, the converted student scores 0.000 at both 1.7B and 8B—not degraded, absent. Diffusion pretraining alone does not bring it back. This is the interference wall our companion paper isolates on a toy bench (Boesch & Wee, 2026), where a table-to-query distance curriculum removes it, so we port that curriculum into the final stage.

The port works, but stochastically; making it reliable is the paper’s main practical result. Under an *open-loop* curriculum, with retrieval distance advancing on a fixed schedule, one seed in three learns to retrieve. Gating the schedule on measured retrieval accuracy, so that distance advances only while the model is keeping up, takes the same three seeds to 3/3, and the resulting recipe carries unchanged to 8B.

Contributions.

1. **A staged conversion pipeline across both axes** (§3), and a stage-attributable account of what transfers: language modeling does; in-context retrieval does not (0.000 at 1.7B and 8B; §4.1).
2. **A closed-loop retrieval curriculum** (§3.3, §4.2). Under

an open-loop distance ramp the repair lands on one seed in three. A success-gated ramp lifts the same seeds to 3/3 at 0.94–0.99 and holds on real text (0.955–0.995 on every evaluated cell).

3. **Transfer to 8B, and the residual failure mode (§4.3).** Recall transfer rises from 0.000 to a mean of 0.901, generalizing to twice the trained pair load and beyond the trained distance. Two of three seeds lock in; the miss is a heavy-tailed bootstrap onset (6.5k, 11k, >16k steps) that a fixed step budget truncates.
4. **A token-coverage boundary (§4.4).** Every model that solves its trained token band reads chance on held-out bands, at both scales. An arm that randomizes key/value roles every batch rules out memorized bindings as the cause; three curriculum policies over randomized pools fail by three distinct mechanisms.
5. **A completed block-diffusion conversion (§5):** a 7B code model into a 3:1 recurrent–attention hybrid, trained to 85,449+4,000 steps divergence-free (held-out NLL 2.819), with its construction and two negative results—a representation-alignment auxiliary that does nothing, and a learning-rate ceiling that appears only late in training.

2. Related Work

AR-to-diffusion adaptation. Gong et al. (2025) and Ye et al. (2025) adapt pretrained autoregressive Transformers into masked diffusion LMs, establishing that the objective change is survivable and far cheaper than pretraining. Both retain attention. **Attention-to-recurrence distillation.** RADLADS (Goldstein et al., 2025) converts attention into RWKV-7-style recurrence at multi-billion scale on a few hundred million tokens, retaining next-token prediction; our stage 1 follows its two-phase recipe. **Recurrent denoisers.** DiffuMamba (Singh et al., 2025) shows that scan backbones are viable denoisers; the rank-1 state update of Peng et al. (2025) and Yang et al. (2024) is the cell we convert into. **Block diffusion** (Arriola et al., 2025) interpolates between the autoregressive and diffusion objectives and is the objective of the conversion in §5. **Recall in recurrence.** Fixed-state recurrences are well documented to underperform attention on associative recall (Arora et al., 2023; 2024). Our companion paper argues on a toy bench that the deficit is substantially a *learnability* problem rather than a capacity limit (Boesch & Wee, 2026), and convergent evidence appears elsewhere (Okpekpe & Orvieto, 2025; Blouir et al., 2024); this paper tests that reading at scale.

3. Method: The Conversion Pipeline

The pipeline takes an attention teacher to an attention-free bidirectional recurrent diffusion student in three stages. Ta-

ble 1 lists the concrete settings; the final stage is configured identically at both scales apart from the corpus and the retrieval-episode probability.

3.1. Stage 1: attention to recurrence

This stage moves the architecture and leaves everything else alone: attention becomes recurrence, while the objective remains next-token prediction. The teacher is a uniform softmax Transformer: Qwen3-1.7B (28 layers) or Qwen3-8B (36 layers, hidden size 4096, 32 query and 8 key–value heads). The student inherits the teacher’s token embeddings, MLPs, normalization layers and output head, all frozen, and replaces *every* attention sublayer with a gated delta-rule recurrent cell (Yang et al., 2024). Training follows RADLADS (Goldstein et al., 2025) in two phases. *Attention transfer* trains each cell in isolation, on the teacher’s own hidden states, to reproduce the output of the attention sublayer it replaces. *Refinement* then trains the assembled student end-to-end to match the teacher’s next-token distribution (logit KL). The teacher is served in 8-bit. The corpus is text8 (95M characters of lowercase Wikipedia text) at sequence length 256; the refinement budget is 60k steps (30.7M tokens) at 1.7B and 120k steps (≈ 30 M tokens) at 8B. We report stage-1 quality as teacher top-1 agreement on held-out text (0.746 at 1.7B, 0.779 at 8B) and perplexity retention (0.932 at 1.7B).

3.2. Stage 2: bidirectional expansion

This stage is structural: it changes the network’s shape without changing its function, and the parameters it adds are learned in stage 3. Each converted layer gains a *backward* cell, initialized as a copy of its forward cell and run over the reversed sequence, plus a merge $W \in \mathbb{R}^{d \times 2d}$ that combines the two streams, initialized to their average. The embedding table gains one row for a mask token. Bidirectionality is what a denoiser requires: every position must see both sides of the sequence. It also gives the student query-first reading for free, because the backward stream reads a query before the text that precedes it.

3.3. Stage 3: diffusion continual pretraining, with a retrieval curriculum

Objective. Masked discrete diffusion (Austin et al., 2021; Sahoo et al., 2024): a fraction of tokens is replaced by the mask token and the model is trained to fill them, with the mask ratio annealed from 0.15 to 0.65 over training. By default the trainable set is the backward cells, the merges and the mask embedding, with the forward cells frozen at their stage-1 values. The `train-fwd` variant also unfreezes the forward cells. Both appear below; the contrast between them is informative (§4.1).

Retrieval episodes. With probability p a training batch is a retrieval episode rather than a text batch. An episode has the layout

$$[\text{text prefix}] [k_1 v_1 \cdots k_N v_N] [\text{text haystack}, g \text{ tokens}] \\ [k_j \text{ MASK}] [k_{j'} \text{ MASK}] \cdots$$

with $N \in \{4, 8, 16\}$ pairs and four queried pairs per episode, every answer masked and supervised. Keys and values are single frequent tokens of the corpus, 64 of each, fixed for the run; the prefix and haystack are real text. The haystack length g , the table-to-query distance or *gap*, is the curriculum variable: each batch samples g uniformly in $[0, \text{cap}]$, and the cap grows from 0 toward the maximum that fits the sequence. Evaluation is always at the largest gaps.

Schedule: open-loop vs. closed-loop. In the *open-loop* (ramp) schedule the cap grows linearly with the step count, reaching its maximum at 80% of training. In the *closed-loop* (success-gated) schedule an exponential moving average (EMA) of retrieval accuracy is maintained over retrieval batches, and the cap advances by a fixed increment only on steps where the EMA is at or above a threshold (0.6). The cap is a ratchet: it never retreats. A *thermostat* variant, used in §4.4, lets the cap retreat while the EMA is below threshold.

Randomized symbol pool. For the coverage experiments of §4.4, keys and values are drawn afresh for every batch from a pool of P tokens ($P \in \{128, 512\}$) rather than a fixed set of 64, so that no particular binding can be memorized. Pool-size *annealing* advances through tiers ($128 \rightarrow 512 \rightarrow 2048$) after a streak of passing steps.

3.4. Evaluation: a teacher-gated recall probe

Retrieval is measured with a multi-query associative recall probe in the episode format above: $N \in \{4, 8, 16\}$ pairs, a real-text haystack of $g \in \{0, 64, 160\}$ tokens, and one masked answer position scored per episode by exact match. The value marginal, ≈ 0.023 , is chance. Two controls accompany every number. *Shuffled*: the bindings are permuted and the original value is scored, so a genuine retriever must collapse to the marginal. *Mismatch*: the query is a key that never appeared, which gives the prior floor. We validated the probe format on the attention teacher before scoring any student: the teacher must clear the marginal by a wide margin and collapse under shuffling. This *teacher gate* rules out probe artifacts as an explanation for student failures.

Trained band and disjoint band. The DISJOINT evaluation is the identical probe with keys and values drawn from a token band that never appears in any retrieval episode during training and is excluded from every randomized pool. It asks whether what was learned is retrieval, or retrieval over particular tokens.

Table 1. Settings at both scales. Stage 3 uses batch 4, sequence length 256, learning rate 5×10^{-4} and 16k steps at both scales. p is the retrieval-episode probability.

	1.7B	8B
teacher	Qwen3-1.7B	Qwen3-8B
student cells	gated delta rule	gated delta rule
stage-1 refinement	60k steps (30.7M tok)	120k steps
stage-1 agreement	0.746	0.779
stage-3 corpus	text8 / wikitext	text8
p (retrieval)	0.7 / 0.5	0.5
gate threshold	0.6	0.6
hardware	RTX 3090, 24 GB	RTX PRO 6000, 96 GB
wall time per arm	≈ 6 h	≈ 2.6 h

Lock-in. In training telemetry, learning to retrieve is a binary event. The retrieval EMA sits near zero for thousands of steps and then rises past 0.6 within roughly a thousand steps; we call that rise the *avalanche*, the step at which it begins the *onset*, and a run that avalanches within its budget *locked in*. Every locked-in run’s evaluation clears both controls. Because the event is binary, we report lock-in rates with per-seed values rather than means.

4. Results

4.1. What transfers, and what does not

Stage 1 transfers language modeling: top-1 agreement with the teacher reaches 0.746 and 0.779 on held-out text. It does not transfer in-context retrieval. On the recall probe the attention teacher scores 0.335–0.350; the converted student scores 0.000 at both 1.7B and 8B, and its accuracy is insensitive to whether the bindings are shuffled—it is emitting a prior, not retrieving. Under stage-3 diffusion pretraining alone (no retrieval episodes) it remains at 0.000. The capacity exists—the same cell trained from scratch on the toy bench solves 16-pair recall at 0.996 (Boesch & Wee, 2026)—but the distillation did not instill it.

The curriculum needs trainable forward cells. Injecting retrieval episodes into stage 3 with the forward cells frozen teaches the model to emit well-formed answers but never to bind: accuracy stays at the prior. With the forward cells unfrozen (`train-fwd`), retrieval is acquired. The mechanism account predicts exactly this: if the circuit is *mark-and-route*—the backward stream marks the binding as it passes the query, the forward stream routes it to the answer position—the routing direction must be trainable. The cost is a small denoising penalty against the no-curriculum control (-0.06 masked-token accuracy).

4.2. The lottery, and the closed-loop fix

Table 2 collects the conversion-scale results, and the pattern is plain: the open-loop curriculum is a lottery, the gated one is not. Under the open-loop ramp at 1.7B, three seeds draw

Table 2. Conversion-scale replication. Retrieval on the trained token band, per seed (mean over $N \in \{4, 8, 16\} \times \text{gap} \in \{0, 64, 160\}$; marginal ≈ 0.023). DISJOINT is the identical probe with key/value symbols from a held-out token band. The open-loop row’s non-zero seed is a partial lock-in (0.40–0.70 across cells); the closed-loop schedule lifts the same three seeds to ceiling. Every row—including every solved one—reads 0.000 on the disjoint band.

setting	lock-in	per-seed (trained)	disj.
1.7B, open-loop	1/3	0.00/0.00/0.55	0.000
1.7B, gated	3/3	0.97/0.98/0.97	0.000
1.7B, gated, wiki	1/1	0.955–0.995	0.000
8B, gated	2/3	0.90/0.78/chance	0.000

$\{0.000, 0.000, 0.40\text{--}0.70\}$ —one partial lock-in in three. An earlier unseeded pilot run of the same configuration had reached 0.94–1.00; an exact rerun returned 0.000, and we count the pilot as one more draw from the same lottery rather than as a result. The same lesson arrived independently on the toy bench, where a two-seed directional margin reversed under ten-seed replication (Boesch & Wee, 2026): in this training regime, low-seed means are not evidence.

Why the open-loop schedule fails. The ramp advances the gap cap on a clock. A run that has not yet learned to bind at short gaps is dragged to long ones regardless, and spends the rest of its budget training at distances it cannot solve. Whether a seed escapes before the ramp outpaces it is close to a coin flip.

The closed-loop schedule. Gating the cap on the retrieval EMA removes the failure: no run is advanced while failing, and a run that stalls simply trains longer where it is. On the same three seeds the gated ramp locks in 3/3, with per-seed means 0.971, 0.977 and 0.974. Accuracy at the maximum evaluated gap matches accuracy at gap zero, and both controls are clean. The telemetry matches the mechanism: all three runs climbed the ramp with the EMA at 0.98–0.99 throughout. Their final caps (0.66–0.87 of the maximum) never reached 1.0, yet evaluation is already at ceiling at the maximum gap; the tail of the ramp is unnecessary at this sequence length.

Real text. On wikitext the gated recipe reaches 0.955–0.995 on every $N \times \text{gap}$ cell, including $N=16$ at gap 160 (0.975), with clean controls. Its telemetry shows why earlier real-corpus attempts had failed: bootstrap is slower on real text (EMA 0.04 at 3k steps, avalanche by 5.4k, a clean ramp to cap 0.91), so open-loop schedules outran it. Real text delays lock-in; it does not prevent it. The conversion-scale recipe is therefore *curriculum + trainable forward cells + success-gated ramp*, and the claim it supports is a lock-in rate.

4.3. Scaling to 8B

Applied unchanged to the 8B student, the recipe lifts recall transfer from 0.000 to 0.69–0.99 (mean 0.901) on the evaluation grid, with a bidirectional denoising gain of +0.117 over the causal stream. A wider evaluation of the same model holds at twice the trained pair load and past the trained distance ($N \leq 32$, gap ≤ 224 : 0.49–0.99, mean 0.786), so both capacity and distance generalize beyond the training distribution.

The residual lottery is bootstrap onset. Across three 8B seeds the tally is 2/3. Every locked-in run, at both scales, shows the same profile: a long dead phase, an avalanche in which retrieval NLL collapses within about a thousand steps, then a clean gated ramp. What varies is *when*. Observed onsets are 6.5k, 11k and >16k steps at 8B, 13k for the randomized-pool arm and 5k for the wikitext arm at 1.7B. The distribution is heavy-tailed, and every run here had a fixed 16k-step budget, so the budget truncates the tail. The amplitudes show the same effect. Seed 1 avalanched at 11k and finished with a weaker floor (worst cell 0.35) than seed 0, which avalanched at 6.5k (worst cell 0.69), because later onset leaves less post-avalanche training. The indicated fix is a success-gated *budget*—train until the avalanche, then a fixed number of steps beyond it—rather than a larger fixed total. It is designed and not yet run; we report 2/3 as it stands.

A second cell family. An RWKV-7 student at 8B trains stably under the same recipe, with a positive bidirectional denoising gain (+0.053), but does not avalanche within 16k steps. Given the onset distribution above we read this as underpowered rather than negative: RWKV-7’s retrieval bootstrap is slower than the gated delta-rule cell’s everywhere we have measured it, and a verdict needs the gated budget.

4.4. What does not transfer: token coverage

Every model that solves the trained-band evaluation—at 1.7B and 8B, including all three gated-ramp seeds and the wikitext arm—scores 0.000 on the DISJOINT band. What the curriculum installs is a binding circuit over the symbols that appeared in retrieval episodes, not an abstract retrieval faculty.

The obvious explanation is memorization of fixed bindings, and it is wrong. An arm trained with keys and values drawn afresh from a 128-token pool every batch, so that no fixed binding *can* be memorized, still locks in (late, at ≈ 13 k steps, with the same avalanche profile), reaches 0.39–0.92 on its trained band, and still reads 0.000 on the disjoint band. It also generalizes in distance far beyond its own training: its cap reached only ≈ 40 tokens, yet gap-64 accuracy matches gap-0 and gap-160 sits well above chance. The circuit is

neither memorizing pairs nor distance-bound. What separates the two bands is which *embeddings* ever participated in a retrieval episode.

Three failure modes, one cause. Randomizing the pool interacts with the gate, and mapping that interaction was necessary to avoid mistaking it for a refutation. At threshold 0.6 the pool task starts out below it, the cap never advances, and the arm trains only at gap ≈ 0 (*lockout*). At threshold 0.25 the cap advances on a weak signal and the EMA then collapses; because the cap is a ratchet, all subsequent training runs at gaps the model cannot solve and the circuit never recovers (*learn-then-collapse*). With the thermostat, a 512-token pool never bootstraps within budget while a 128-token pool does, at 13k steps (*non-bootstrap*). Three policies, three mechanisms, one cause: the randomized-pool task has a much later bootstrap onset, and every policy we tried was budget-limited rather than policy-limited.

Pool annealing is untested. The diagnosis points at pool-size annealing: ride the avalanche on a narrow pool, then widen coverage. At 8B the one annealing run never bootstrapped its first tier inside a 24k-step budget, so the tiers never engaged. Annealing is therefore untested rather than refuted, and it is blocked on the same success-gated budget that §4.3 identifies.

A hybrid alternative. Because token coverage resisted every training-side intervention we tried, retaining a sparse set of attention layers is worth naming as a third reason for hybrids, alongside capacity and length. An attention-retaining variant of the pipeline (keeping every fourth attention layer) passes end-to-end validation on a small teacher, and §5 uses it at 7B; its retrieval has not yet been measured.

5. Block-Diffusion Conversion of a 7B Code Model

The conversions above use full-sequence masked diffusion. A block-diffusion objective (Arriola et al., 2025), which denoises within fixed-size blocks conditioned on committed earlier blocks, interpolates toward autoregression and is a better match for an instruction-following target. We converted a Qwen2.5-Coder-7B teacher under it, and the run completed cleanly: 85,449 pretraining steps (700M tokens) followed by 4,000 response-masked SFT steps, both phases divergence-free under the guard stack.

Outcome. Held-out block-diffusion NLL is 2.819 (masked accuracy 0.512; the training-objective EMA closed at 2.786, a 0.03-nat generalization gap), against a teacher-relative clean-stream AR ratio of $5.4\times$ (student 2.96 vs. teacher 1.27) on fresh documents. The conversion is therefore a working block-diffusion LM with a measured language-modeling floor. The retrieval curriculum was not part of this leg, and the AR ratio says the hybrid still models noticeably

worse than its teacher on the causal stream. Generation-quality evaluation, instruction tuning, and decoding for this model are the subject of a forthcoming results paper; here we report its construction and training health, which differ from the pipeline above in ways that matter.

Architecture. The teacher has 28 layers (hidden size 3584; 28 query and 4 key-value heads). The student replaces 21 of its attention layers with gated delta-rule cells and keeps every fourth attention layer as-is, a 3:1 recurrent-attention hybrid. The cells are teacher-matched: 28 heads of key dimension 128 and value dimension 256, with the query and output projections copied directly from the replaced attention layer and the key/value projections tiled from its grouped-query heads. The kept attention layers are trainable and initialized as exact copies of the teacher’s. Embeddings, MLPs, norms and the output head are inherited; MLPs, norms and the head stay frozen. Trainable parameters total 2.91B. Stage 1 is as in §3.1, with the copied initialization replacing random initialization and refinement stopped on an agreement plateau. Stage 2 is not needed: block diffusion is bidirectional only within a block, and the within-block context is provided by the noised stream described next rather than by a backward recurrence.

Two-stream training. Each step builds a clean and a noised view of the same batch. The clean stream is an ordinary causal pass of the network. The noised stream conditions, per 64-token block, on the clean stream’s state for strictly-previous blocks and on itself within the block. Setting the diffusion block equal to the recurrent chunk size makes this nearly free: the chunked recurrence already materializes each chunk’s incoming state, so exposing it yields “clean recurrent state at the start of block n ” at no extra forward cost, and the noised stream needs no scan of its own. In the noised pass, the kept attention layers attend to clean keys from previous blocks and noised keys within the block. Per-block noise levels are drawn uniformly on $[0.2, 1]$ with the usual $1/t$ weighting, and the output is shifted by one position so that the teacher’s next-token head semantics are preserved.

Budget. Sequence length 512, 8192 tokens per step, 85,449 pretraining steps (700M tokens from a 2.1B-token code-and-text corpus) on a single TPU v5e-8 host, followed by a 4,000-step supervised phase that denoises only the response region of chat-formatted examples (512×16 , response mask ratio $\sim U[0.3, 1.0]$, ten-percent corpus regularizer).

Negative result 1: a representation-alignment auxiliary does nothing. We added a term aligning the student’s clean-stream hidden states to the frozen teacher’s at each layer boundary, expecting it to preserve teacher structure through the objective change. Measurement showed otherwise. Cosine alignment was already $\approx 94\%$ saturated at initialization, because the copied projections and frozen backbone anchor

the hidden states on their own. The term did not improve autoregressive retention, inflated logit scale roughly twofold, and cost 10–15% of step time. It is off.

Negative result 2: the learning-rate ceiling appears late.

At learning rate 3×10^{-4} the run diverged twice, both times deep into training (around steps 50–52k and 56k) and with the same signature: escalating oscillation in loss over several hundred steps, then spike, recover, collapse, with accuracy falling to ≈ 0.02 . No single step exceeded the loss-spike guard’s threshold, so the step-level guard registered nothing. Training at 2×10^{-4} is stable, as it was for the 1.5B validation run that preceded this one. The failure is invisible to the obvious monitor and appears only after tens of thousands of stable steps.

6. Limitations

1. **Token generality is unsolved.** Every solved model reads 0.000 on held-out token bands (§4.4). We make no claim that the curriculum installs a general retrieval faculty; the annealing fix that our own diagnosis motivates is untested.
2. **Three seeds per setting.** Conversion-scale rows rest on three seeds (one for the wikitext arm). 3/3, 2/3 and 1/1 are small- n rates, and no pairwise contrast in this paper is significance-tested.
3. **Fixed budgets confound several negatives.** The 8B seed-2 miss, the RWKV-7 non-bootstrap and the pool-annealing arm are all consistent with budget truncation of a heavy-tailed onset rather than method failure. Until the success-gated budget is run, those three are open.
4. **One probe family.** Retrieval is measured with a single multi-query recall probe whose format is adjacent to the training episodes. Held-out bindings and a held-out token band are used, but the format proximity is real, and we do not claim broad downstream retrieval.
5. **The block-diffusion conversion is reported as construction and training health only (§5):** training finished cleanly with a measured holdout floor, and the $5.4 \times$ AR ratio marks a real gap to the teacher; generation, retrieval transfer, and instruction-following evaluation are deferred to a forthcoming results paper.

7. Discussion

What conversion costs. Teacher knowledge survives crossing both axes; the in-sequence binding faculty survives neither. It is absent rather than degraded, and diffusion continual pretraining does not bring it back. It has to be re-trained deliberately, and the toy-bench mechanism account (Boesch & Wee, 2026) says why: the wall is a training-coverage gap, so only training that covers the gap can close

it.

Closed-loop curricula are the transferable lesson. An open-loop curriculum is a clock; a closed-loop one is a controller; the difference is 1/3 versus 3/3 on identical seeds. The same distinction predicts the toy bench’s length boundary, where a time-based ramp collapses to 0/6 at $L=512$ (Boesch & Wee, 2026). Wherever a curriculum has a difficulty knob and a measurable competence signal, gating the former on the latter converts a lottery into a method. The natural extension, and the one all of our open negatives point at, is to gate the training *budget* the same way: a fixed step count is itself an open-loop decision about a heavy-tailed random variable.

What the coverage boundary means. A converted model whose retrieval was trained over a narrow symbol pool reads chance outside it, and that boundary survived pool randomization, three gate policies and a scale change. Curriculum-installed retrieval is a circuit over participating embeddings, which makes token coverage a first-class axis of the training distribution. It also supplies a third reason to keep some attention in a hybrid, alongside capacity and length.

Reproducibility Statement

Per-arm training telemetry (gate cap, retrieval EMA, avalanche onset), evaluation outputs for every arm including the failed ones, and checkpoints for every arm except the unseeded pilot are retained in object storage. The toy-bench code and results referenced throughout are public at <https://github.com/JIBSIL/dualgoose>.

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Impact Statement

This work concerns making pretrained language models cheaper to re-architect, which lowers the compute cost of deploying capable models and correspondingly lowers the barrier to deploying them without scrutiny. The retrieval boundary we report is also safety-relevant: a converted model can look competent on its training distribution and read at chance just outside it, a failure mode that perplexity-shaped evaluation does not surface.

References

Arora, S., Eyuboglu, S., Timalsina, A., Johnson, I., Poli, M., Zou, J., Rudra, A., and Ré, C. Zoology: Measuring and improving recall in efficient language models. *arXiv*

- preprint arXiv:2312.04927*, 2023.
- Arora, S., Eyuboglu, S., Zhang, M., Timalsina, A., Alberti, S., Zinsley, D., Zou, J., Rudra, A., and Ré, C. Simple linear attention language models balance the recall-throughput tradeoff. *arXiv preprint arXiv:2402.18668*, 2024.
- Arriola, M., Gokaslan, A., Chiu, J. T., Yang, Z., Qi, Z., Han, J., Sahoo, S. S., and Kuleshov, V. Block diffusion: Interpolating between autoregressive and diffusion language models. *arXiv preprint arXiv:2503.09573*, 2025.
- Austin, J., Johnson, D. D., Ho, J., Tarlow, D., and van den Berg, R. Structured denoising diffusion models in discrete state-spaces. *arXiv preprint arXiv:2107.03006*, 2021.
- Blouir, S., Smith, J. T., and Anastasopoulos, A. Birdie: Advancing state space models with reward-driven objectives and curricula. In *Proceedings of EMNLP*, 2024.
- Boesch, J. and Wee, A. Anatomy of associative recall in fixed-state recurrences: A matched-state decomposition, an interference wall, and a curriculum that breaks it. Companion preprint, 2026.
- Goldstein, D., Alcaide, E., Lu, J., and Cheah, E. RADLADS: Rapid attention distillation to linear attention decoders at scale. *arXiv preprint arXiv:2505.03005*, 2025.
- Gong, S., Agarwal, S., Zhang, Y., Ye, J., Zheng, L., Li, M., An, C., Zhao, P., Bi, W., Han, J., Peng, H., and Kong, L. Scaling diffusion language models via adaptation from autoregressive models. *arXiv preprint arXiv:2410.17891*, 2025.
- McCartney, G., Hacker, T., and Yang, B. Empowering Faculty: A Campus Cyberinfrastructure Strategy for Research Communities. *Educause Review*, 2014.
- Okpekpe, D. and Orvieto, A. Revisiting associative recall in modern recurrent models. *arXiv preprint arXiv:2508.19029*, 2025.
- Peng, B., Zhang, R., Goldstein, D., et al. RWKV-7 “Goose” with expressive dynamic state evolution. *arXiv preprint arXiv:2503.14456*, 2025.
- Sahoo, S. S., Arriola, M., Schiff, Y., Gokaslan, A., Marroquin, E., Chiu, J. T., Rush, A., and Kuleshov, V. Simple and effective masked diffusion language models. *arXiv preprint arXiv:2406.07524*, 2024.
- Singh, V., Ostapenko, O., Noël, P.-A., Belilovsky, E., and Scholak, T. DiffuMamba: High-throughput diffusion LMs with Mamba backbone. *arXiv preprint arXiv:2511.15927*, 2025.
- Yang, S., Kautz, J., and Hatamizadeh, A. Gated delta networks: Improving Mamba2 with delta rule. *arXiv preprint arXiv:2412.06464*, 2024.
- Ye, J., Xie, Z., Zheng, L., Gao, J., Wu, Z., Jiang, X., Li, Z., and Kong, L. Dream 7b: Diffusion large language models. *arXiv preprint arXiv:2508.15487*, 2025.